

# Mohammad Ninad Mahmud Nobo

Bangladesh University of Engineering and Technology (BUET) | CSE

Email: [mninadmnobo@gmail.com](mailto:mninadmnobo@gmail.com)    GitHub: [github.com/mninadmnobo](https://github.com/mninadmnobo)    LinkedIn: [linkedin.com/in/mninadmnobo](https://linkedin.com/in/mninadmnobo)  
Portfolio: [mninadmnobo.github.io](https://mninadmnobo.github.io)    Scholar: [Google Scholar \(M Ninad M Nobo\)](#)    Phone: +880 1939-444451

## SUMMARY

Experienced in Artificial Intelligence, Machine Learning, Large Language Models, and software engineering across research and industry. Skilled in building production-grade software and AI systems, including deep learning models, multi-agent AI pipelines, and full-stack web applications. Research background spans LLM-based software testing, medical AI and applied machine learning.

## EDUCATION

**Bangladesh University of Engineering and Technology (BUET)** **2022 – 2026**  
Bachelor of Science in Computer Science and Engineering (CSE) **CGPA: 3.60 / 4.00**  
**Relevant Coursework:** Data Structures and Algorithms, Algorithm Engineering, Database Systems, Operating Systems, Computer Architecture, Computer Networks, Compiler Design, Artificial Intelligence, Machine Learning, Computer Graphics  
**Research Interests:** Large Language Model Systems, AI-Assisted Software Engineering, Applied Machine Learning, Medical AI

## PROFESSIONAL EXPERIENCE

- **AI Software Engineer** [ [R&D Website](#) | [FABINS](#) ] 2026 – Present  
*Department of Research and Development (R&D) - Saturn Textiles Limited*
  - Developing and training computer vision and deep learning models for automated fabric inspection through **FABINS**.
  - Developed full-stack applications with Next.js, React, Spring Boot, and REST APIs, with GitHub Actions for CI/CD.

## RESEARCH EXPERIENCE

- **AutoTestGenX: Multi-Agent LLM Framework for End-to-End Web Testing (Undergraduate Thesis)** [ [GitHub](#) ] 2025 – 2026  
*Python • LLM Engineering • Multi-Agent Systems • Web Testing • Browser Automation*
  - Designed a multi-agent LLM framework to automate web test generation and execution from natural-language requirements.
  - Built benchmark datasets and evaluated against ground-truth suites, achieving 84.0% scenario coverage and 90% error detection.
- **MedCAR: Conflict-Aware Medical Reasoning System (Under Review)** [ [GitHub](#) ] 2026  
*Python • Deep Learning • Medical AI*
  - Built a multi-model chest X-ray reasoning system with a conflict-resolution pipeline to reconcile contradictory model outputs.
  - Designed a framework using confidence calibration and uncertainty-based abstention for reliable predictions.
- **Bengali-Loop: Community Benchmarks for Long-Form Bangla ASR and Speaker Diarization** [ [arXiv: 2602.14291](#) ] 2026  
*Python • Speech Processing • Machine Learning*
  - Supported data collection and annotation for a large-scale Bangla Automatic Speech Recognition (ASR) and speaker diarization benchmark, enabling evaluation using Word Error Rate (WER) and Diarization Error Rate (DER).

## ACADEMIC PROJECTS

- **MindTrace: AI-Powered Dementia Care Application** [ [GitHub](#) | [Feature](#) | [Infrastructure](#) ] 2025  
*Kotlin • Spring Boot • Spring AI • PostgreSQL • Redis • Firebase • Docker • Azure*
  - Engineered an AI-driven Android app for dementia care with an Android frontend and scalable Spring Boot architecture.
  - Designed accessibility-first UI/UX for cognitively impaired users to reduce cognitive load and improve engagement.
- **Gemma VetCare: AI-Assisted Farming Support System** [ [GitHub](#) | [Feature](#) ] 2025  
*Kotlin • Android Studio • Spring Boot • Spring AI • MongoDB*
  - Built an AI-assisted Android application delivering real-time veterinary diagnosis and treatment recommendations.
  - Designed RESTful APIs optimized for low-connectivity environments with efficient request handling.
- **TCP Window Scaling Attack** [ [GitHub](#) ] 2025  
*Python • TCP/IP • Network Security • Wireshark • Linux*
  - Built a virtualized MITM lab to demonstrate ARP poisoning and TCP window scaling attacks between client-server VMs.
  - Implemented attack and defense clients and analyzed TCP traffic using Wireshark and packet captures.
- **Compiler Construction (Flex, Bison, 8086 Codegen)** [ [GitHub](#) ] 2024  
*C++17 • Flex • Bison/Yacc • Linux*
  - Built a full compiler pipeline for a C-like language, covering lexical analysis, parsing, and semantic analysis.
  - Generated optimized 8086 assembly code with symbol table management and basic optimizations.
- **Smart Gardening Automation** [ [GitHub](#) | [Feature](#) ] 2024  
*Arduino • Embedded C++ • Sensors • Bluetooth Communication • Automation*
  - Developed a gardening automation system integrating environmental sensing and wireless communication.
  - Implemented automated watering, fertilizer/pesticide cycles, smart shade control, and remote monitoring.

## TECHNICAL SKILLS

**AI/ML:** PyTorch, TensorFlow, Scikit-learn, Large Language Models, Prompt Engineering, Multi-Agent Systems, Computer Vision  
**Languages:** Python, C/C++, Java, JavaScript, Kotlin, SQL  
**Backend & Full-Stack:** Spring Boot, Spring AI, Node.js, Express.js, Next.js, REST APIs, React, Android  
**Databases:** PostgreSQL, MongoDB, Redis  
**DevOps & Tools:** Linux, Git, Docker, Docker Compose, GitHub Actions, Firebase, Azure

## ADDITIONAL INFORMATION

**Spoken Languages:** Bangla (Native), English (Fluent), Hindi & Urdu (Conversational), Arabic (Reading)  
**Programming Profiles:** [NeetCode: mninadmnobo](#)    [LeetCode: mninadmnobo](#)  
**Research Profiles:** [ORCID: 0009-0006-2781-6693](#)    [ResearchGate: Mohammad Ninad Mahmud Nobo](#)